Plasma Spectroscopy

(Spectroscopic Study of Gas Discharges)

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Outline

- What is a plasma?
- Plasma applications and why do we study plasmas?
- Origin of light emission (the teacher will ask you several questions regarding light emission!)
- What can optical emission spectroscopy tell us? The birth of quantum mechanics!
- What is a spectrometer?
- Recent research results using OES in plasmas

What Is Plasma

- A plasma is a ionized gas with equal numbers of positive and negative charges.

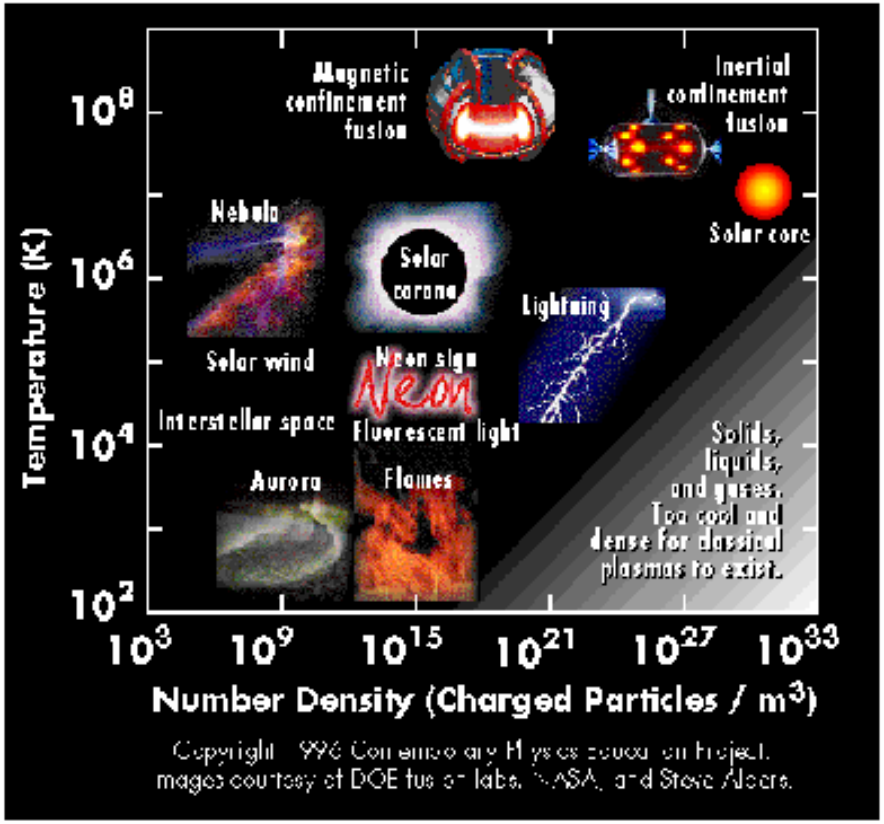
- A more precise definition: *a plasma is a quasi-neutral gas of charged and neutral particles which exhibits collective behavior*.

- Examples:

Sun, flame, neon light, etc.

Why Study Plasma Spectroscopy?

Examples of Plasma

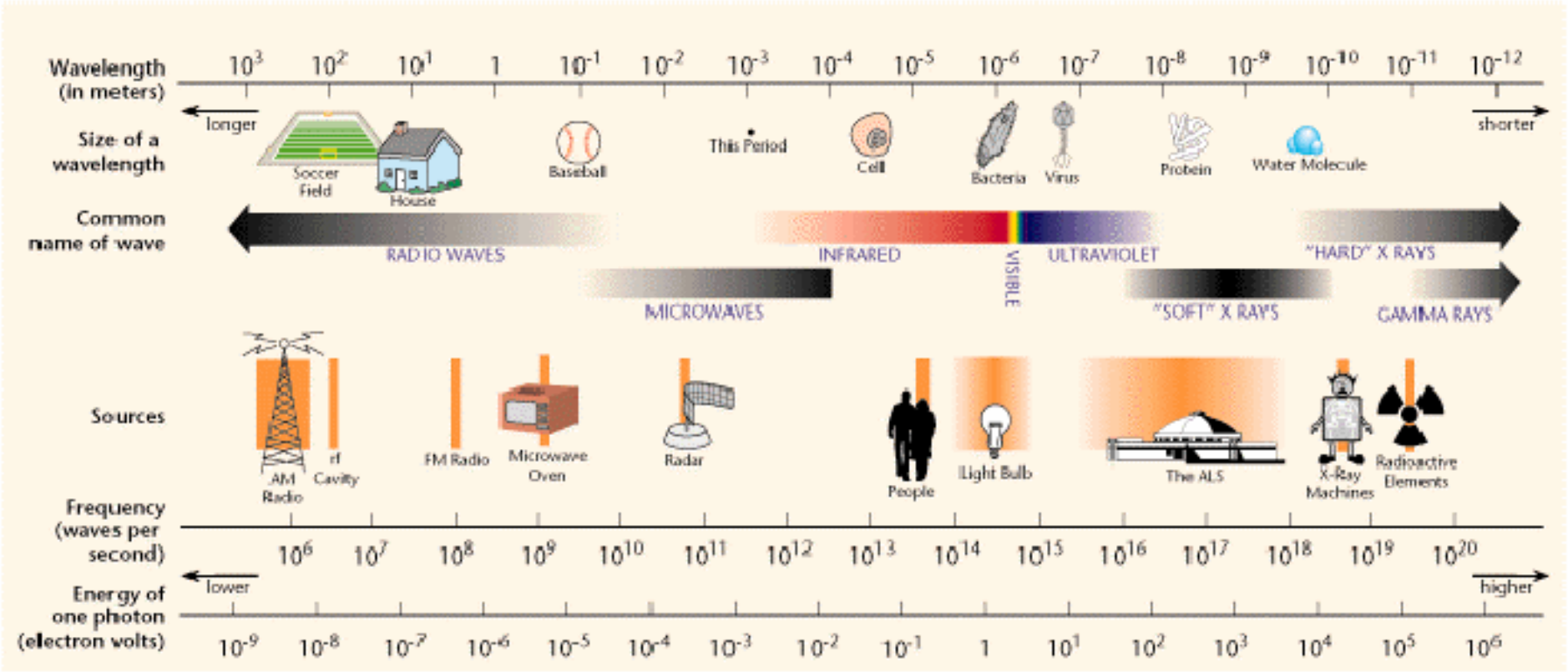


Common Aspect:
Emission of
Photons

↓

Study of Radiation
from Plasma

The electromagnetic spectrum

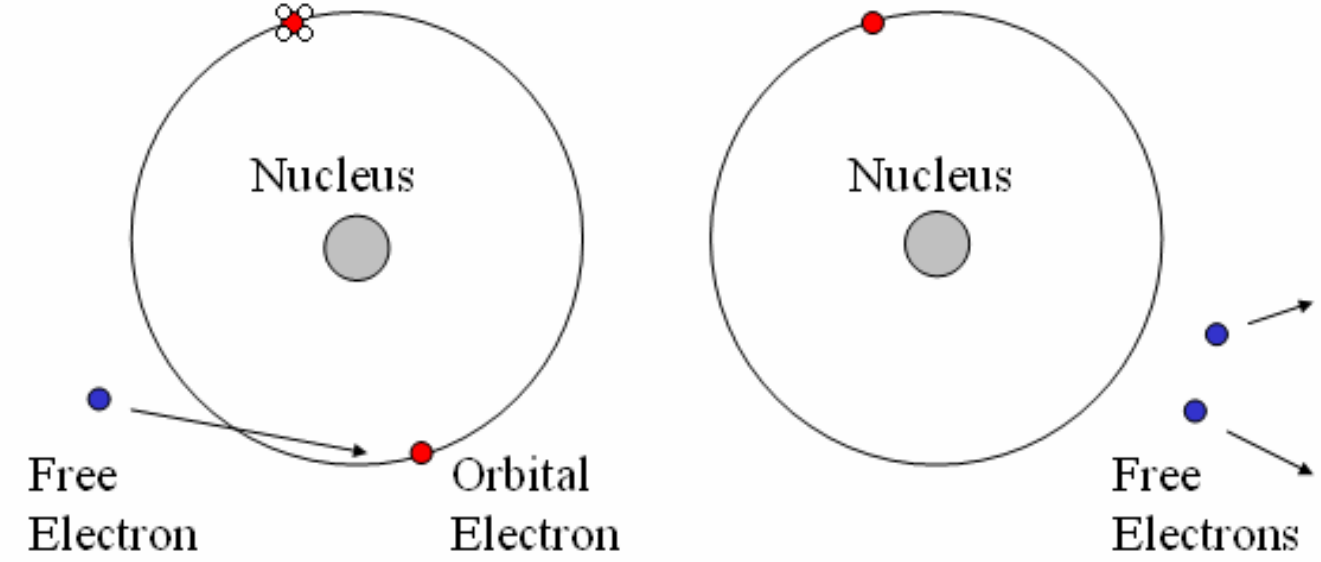


Ionization

- Electron collides with neutral atom or molecule
- Knock out one of orbital electron

$$e + A \longrightarrow A^{+} + 2\,e$$

- Ionization collisions generate electrons and ions
- It



Excitation-Relaxation

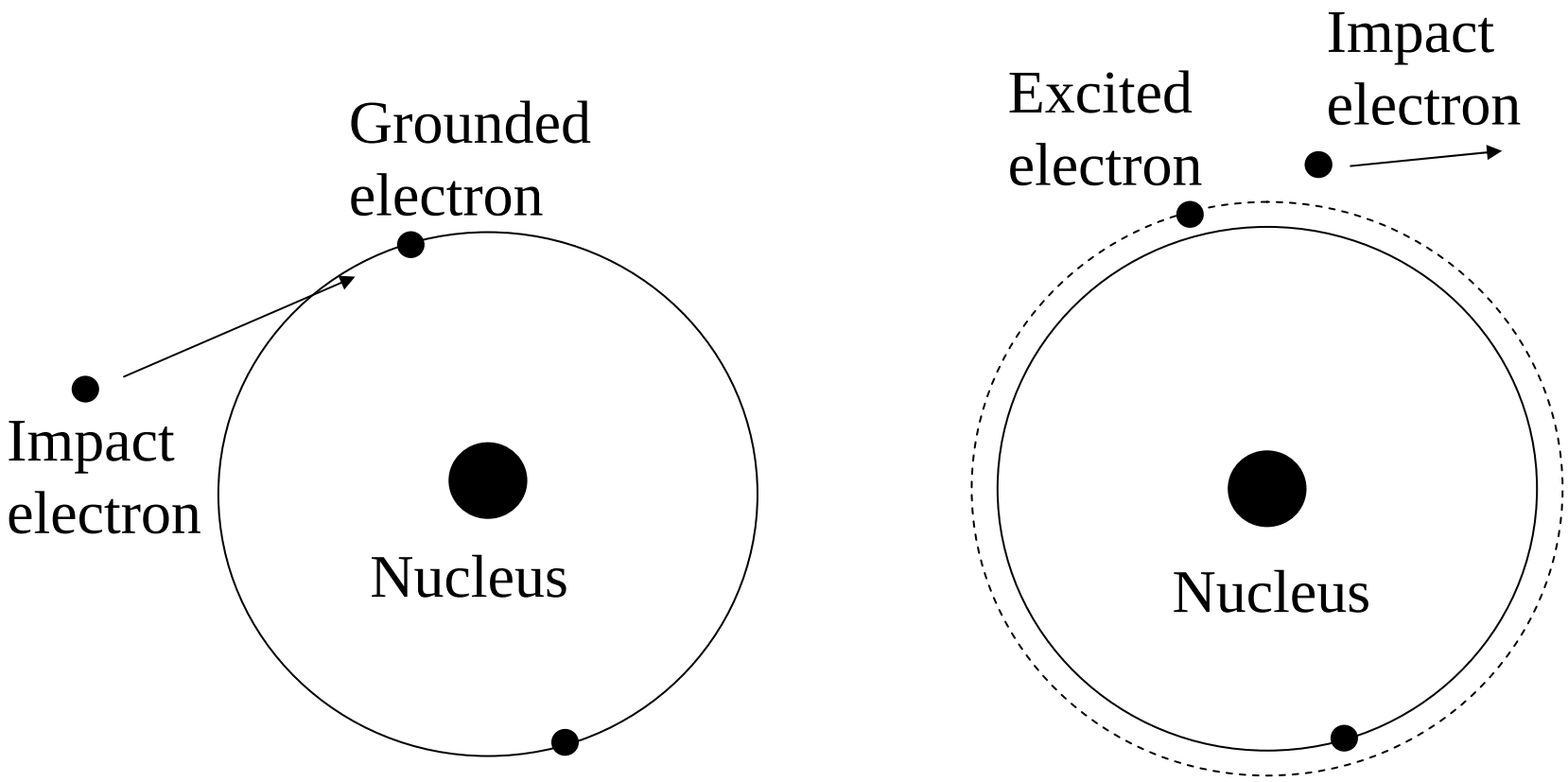
$$e + A \longrightarrow A^* + e$$

$$A^* \longrightarrow A + h\nu\,(\text{Photos})$$

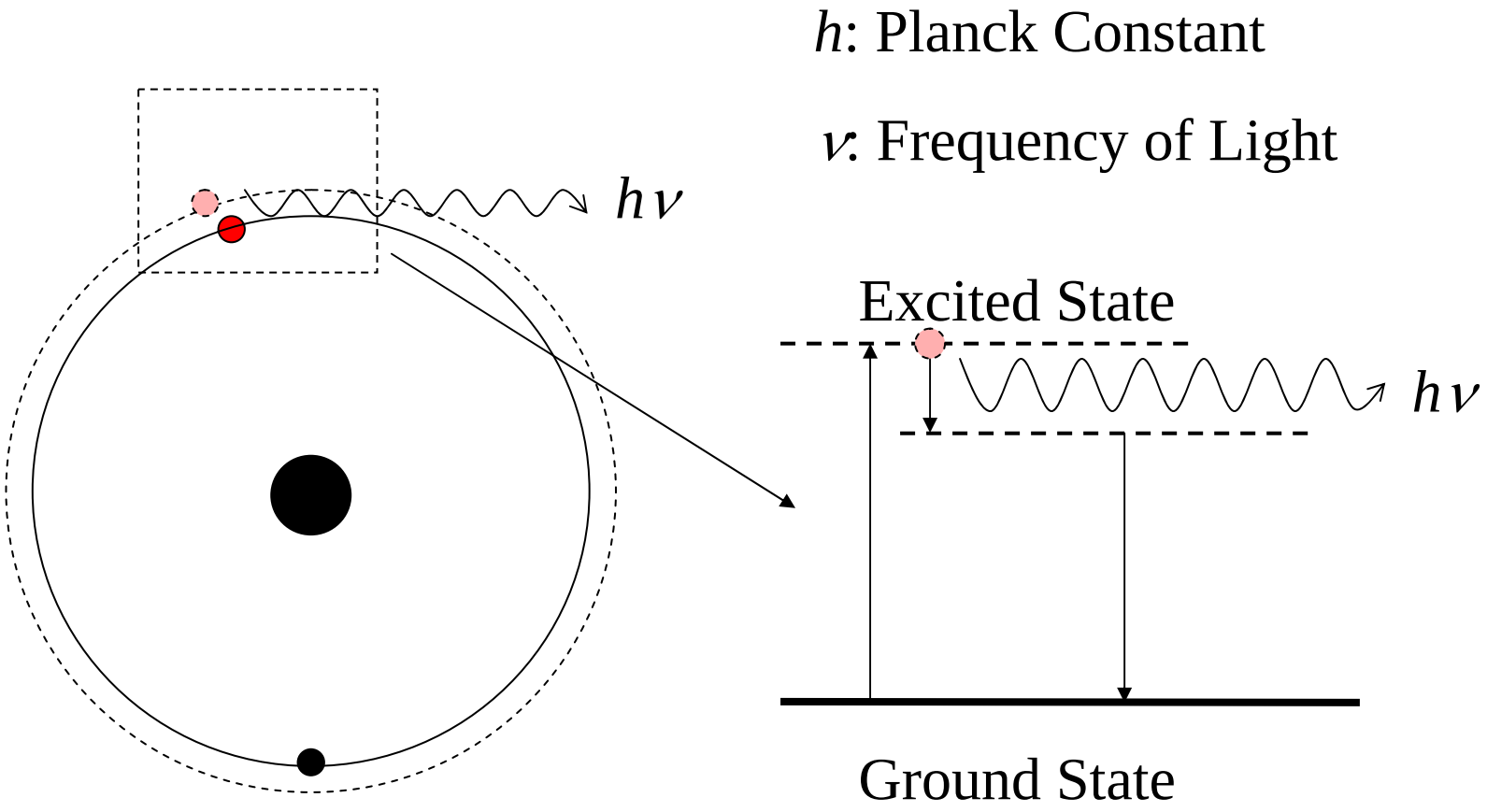
- Different atoms or molecules have difference frequencies, that is why different gases have different glow colors.

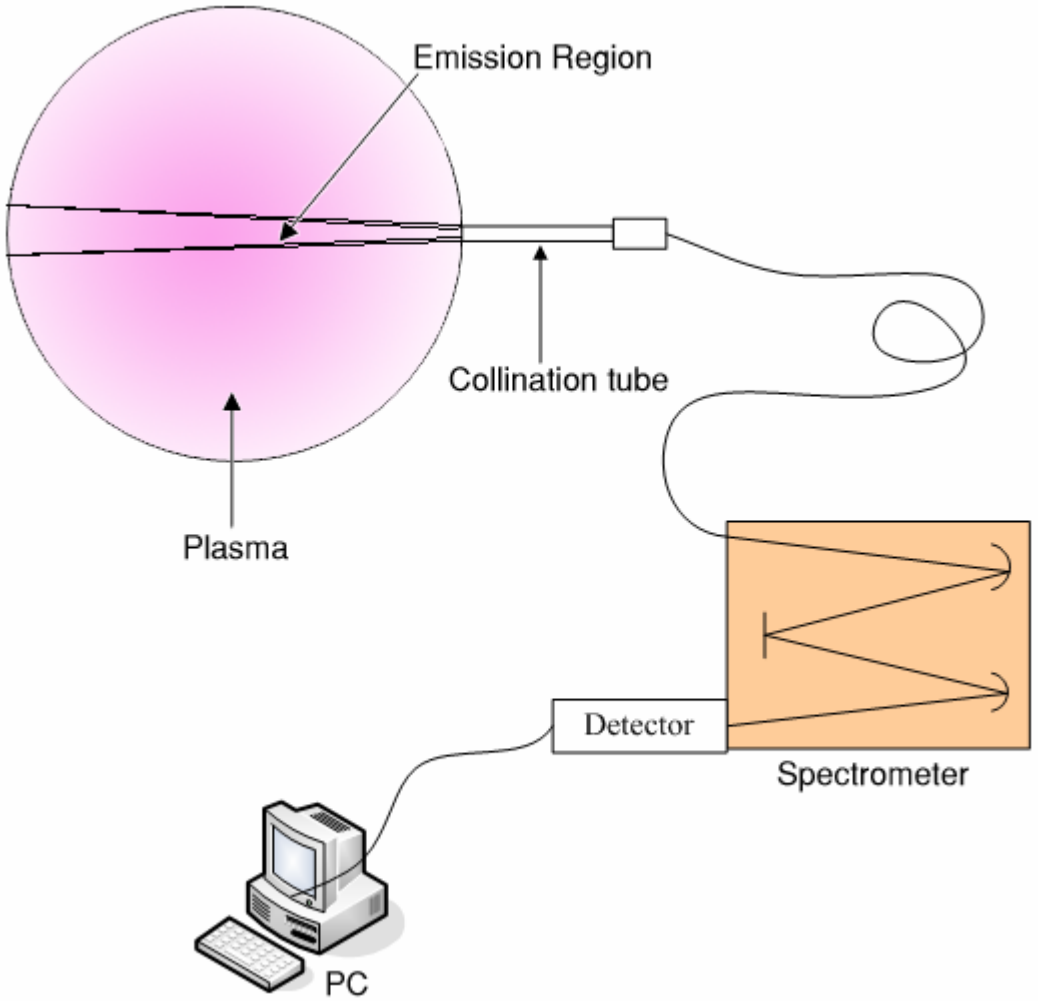
- The change of the glow colors is used for etch and chamber clean process **endpoint**.

Excitation Collision



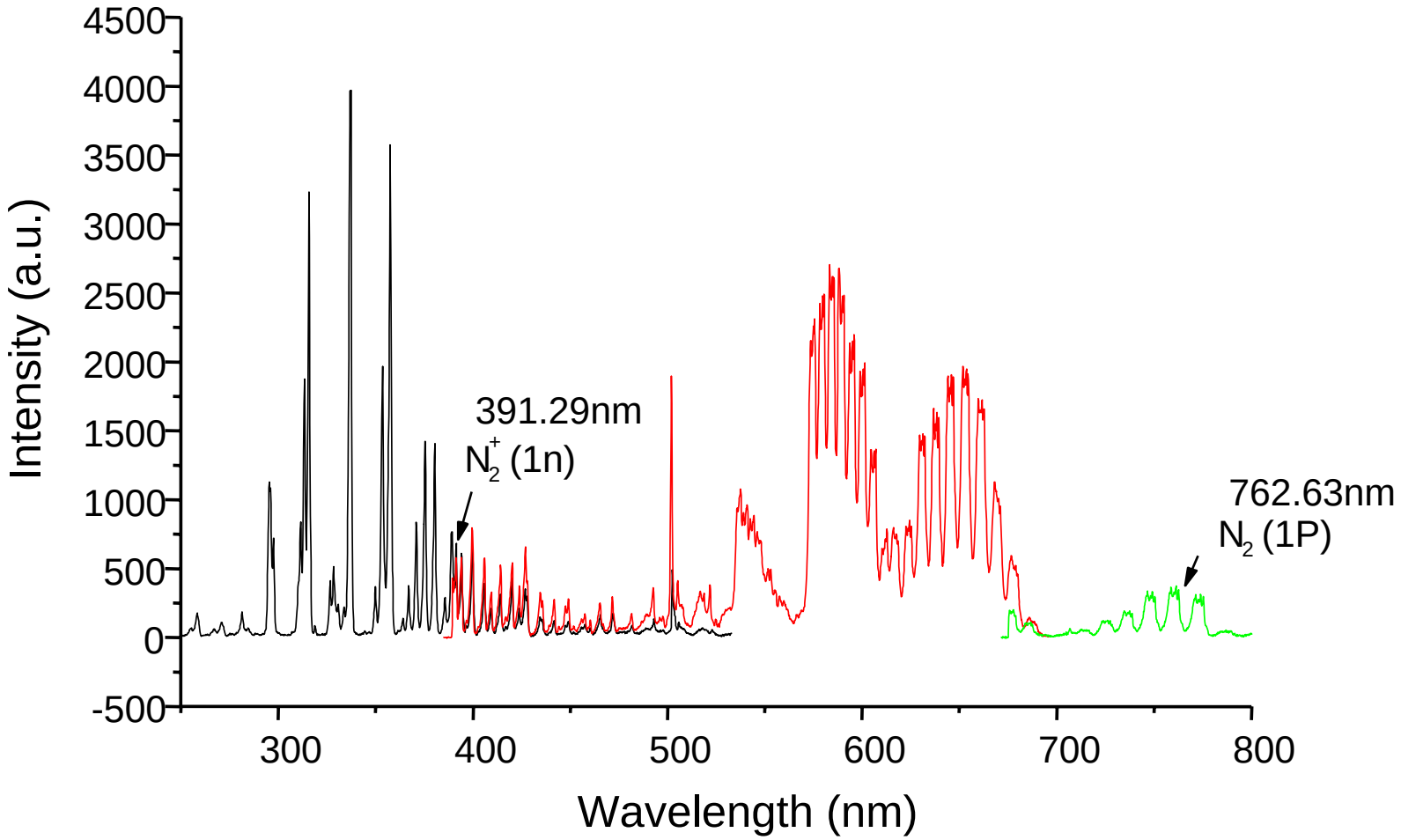
Relaxation


$h\nu$
h: Planck Constant
ν: Frequency of Light
Excited State
$h\nu$
Ground State


Emission Region
Collination tube
Plasma
Detector
Spectrometer
PC

典型的氮等离子体光谱

$$I_1/I_2 = (A_1/A_2)(<\sigma v>_1 / <\sigma v>_2)$$
$$= A_1/A_2\, f(T_e)$$



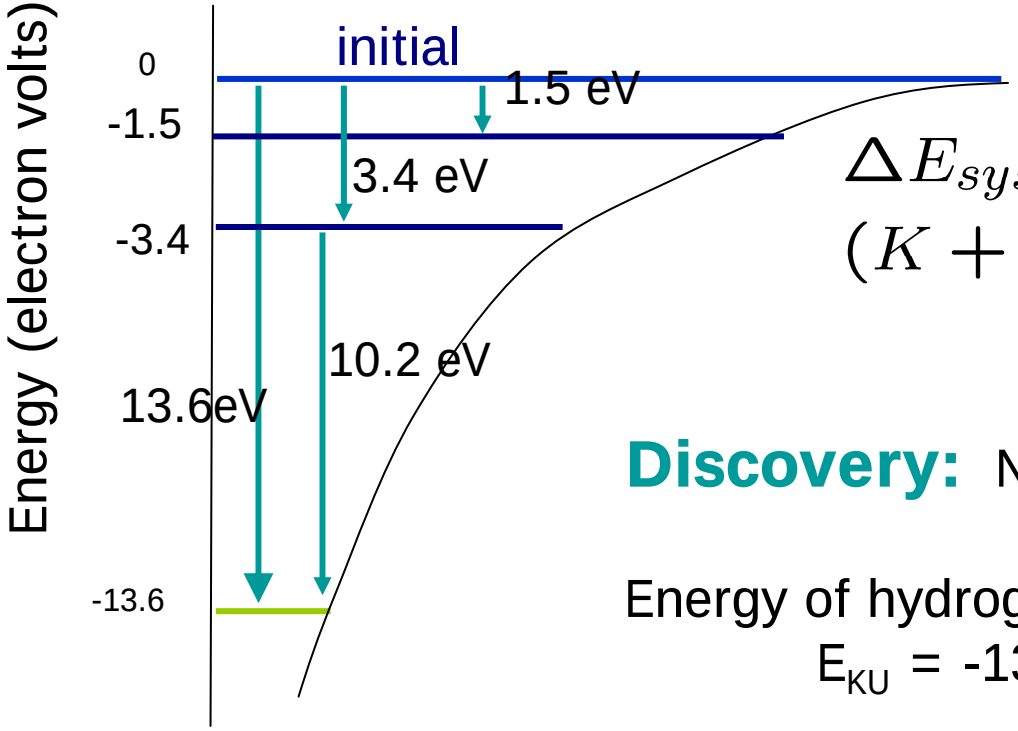
Birth of Quantum Mechanics

Experiment: Very slow electrons encounter protons

They form a hydrogen atom..

Energy of photon gives E_{KU} of Hydrogen Atom

e + p ! H+ γ



$$\Delta E_{sys} = W_{ext}$$

$$(K+U)_{ep-final} - 0 = -E_{\gamma}$$

Discovery: Not all values of (K+U) are seen!

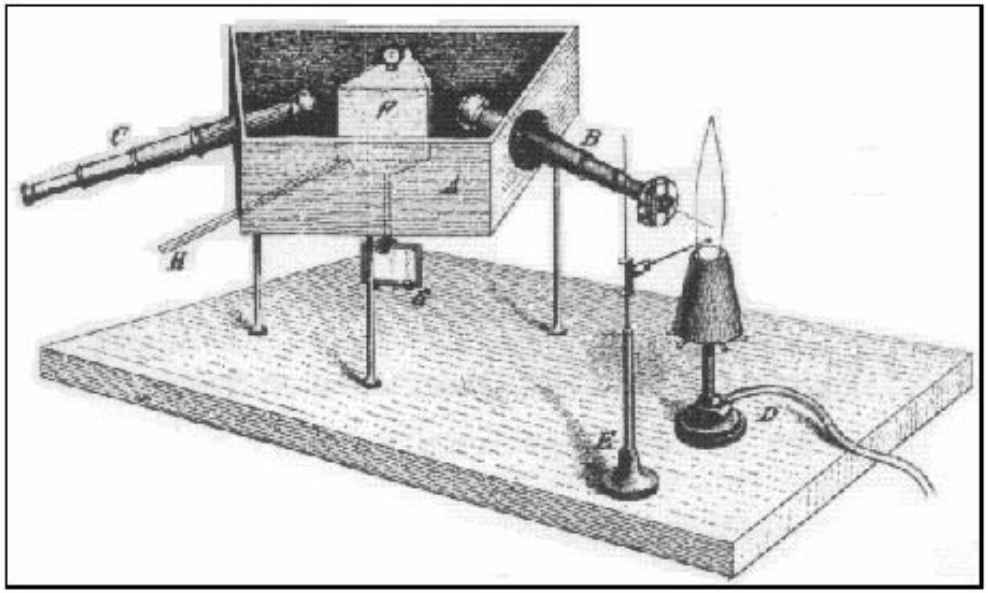
Energy of hydrogen atom given by

E_{KU} = -13.6 eV /N^2 N=1,2,3,...

The oldest spectrometer

Prism spectrograph

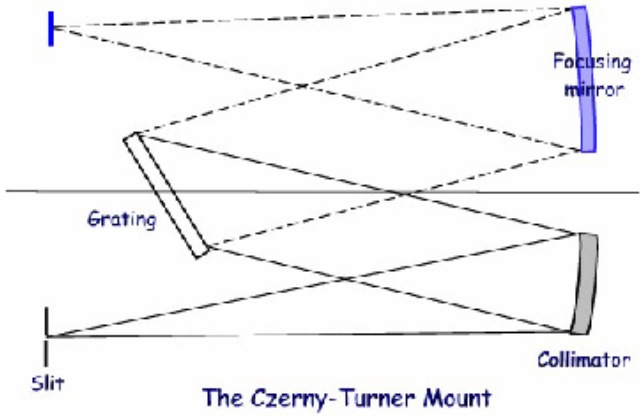






GUSTAV KIRCHHOFF AND ROBERT BUNSEN
Annalen der Physik und der Chemie (Poggendorff), Vol. 110 (1860), pp. 161-189 (dated Heidelberg, 1860)

Grating spectrometer (Czerny-Turner mounting)



The Czerny-Turner Mount

Light from an appropriate source is focussed onto an input slit, and light passing through this slit is collimated by a concave mirror, which also reflects it onto a diffraction grating.

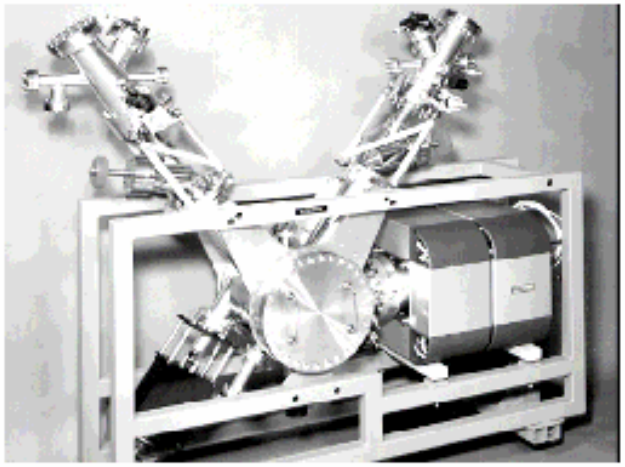
The grating directs the light onto a second concave mirror, which reflects and focusses it onto an exit slit.

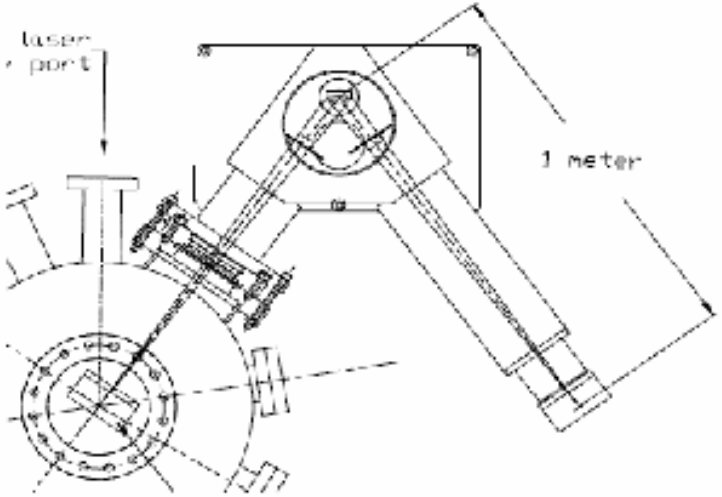
Mirrors are used because they do not introduce any chromatic aberration, but other aberrations limit the resolution of the instrument.

Seya-Namioka VUV spectrometer

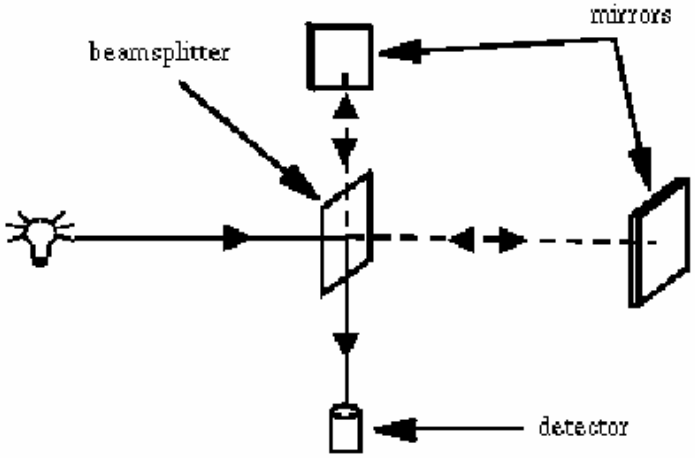
The Seya-Namioka mounting is used at wavelengths > 30 nm and is a special case of the Rowland mounting with an acceptable spectral resolution. It is widely used in plasma diagnostics and at synchrotrons.







Fourier transform spectrometer



A Fourier transform spectrometer is an adaption of the Michelson interferometer.

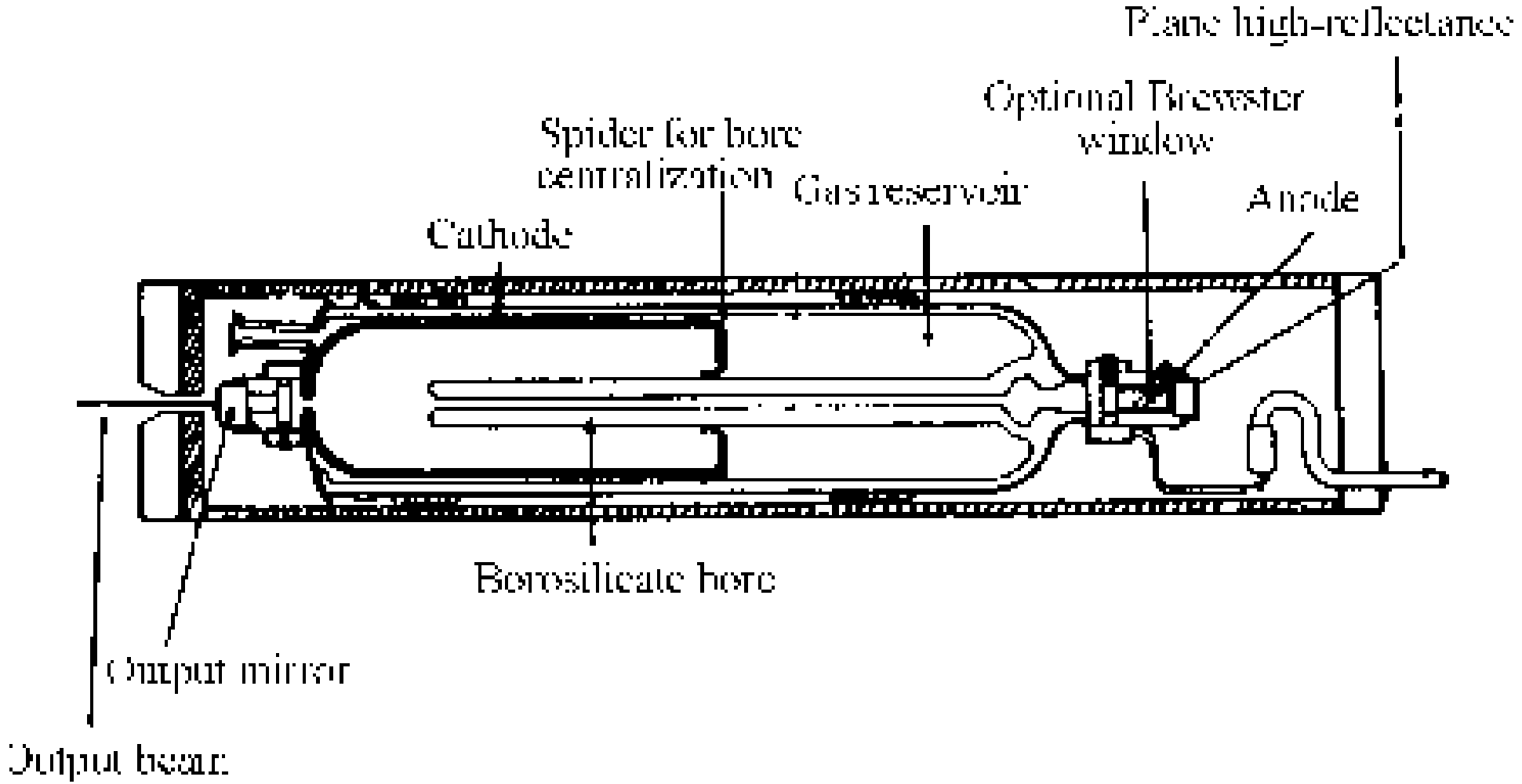
A collimated beam from a light source is divided into two by a beamsplitter and sent to two mirrors. which reflect the beams back along the same paths to the beamsplitter, where they interfere.

The signal recorded at the output depends on the wavelength of the light and the optical path difference between the beamsplitter and each of the two mirrors.

He-Ne Plasmas

- He-Ne Laser is a good example of applications of plasma emissions
- Understanding He-Ne laser operation involves knowledge in quantum mechanics (atomic physics), gas discharge physics and optics

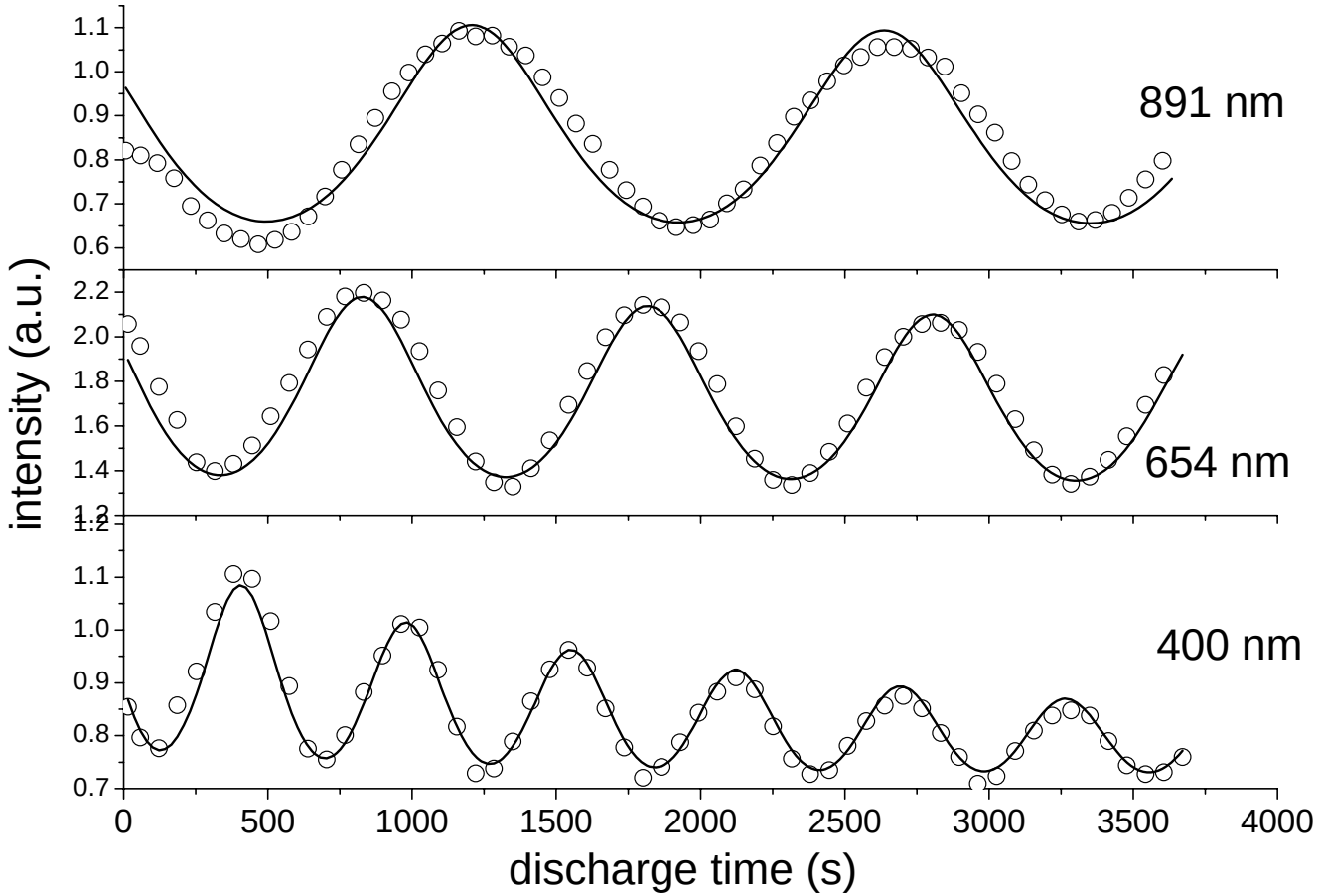
He-Ne Laser Configuration



Research Activities in Our Group

- OES to determine electron temperature
- Absorption spectroscopy to determine metastable densities
- OES to determine mode transitions in plasmas
- OES to determine electron density
- OES to determine film thickness

Optical Interference in Plasmas



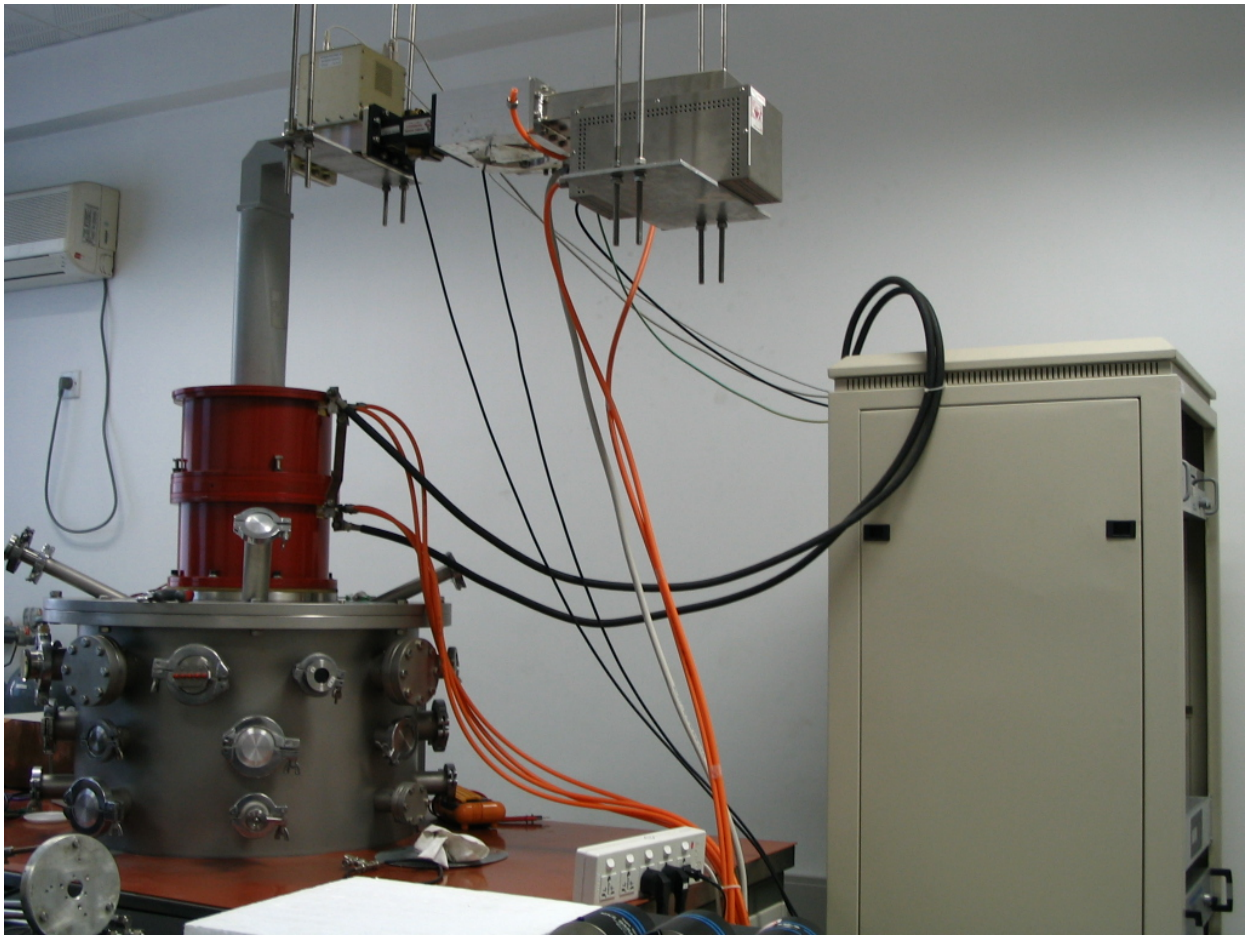
The following are some of the equipment in the lab

- Inductively coupled plasma reactor
- Electron Cyclotron Resonance plasma reactor
- Capacitively coupled plasma reactor
- Four spectrometers (of total seven) in the lab

ICP Plasma Reactor



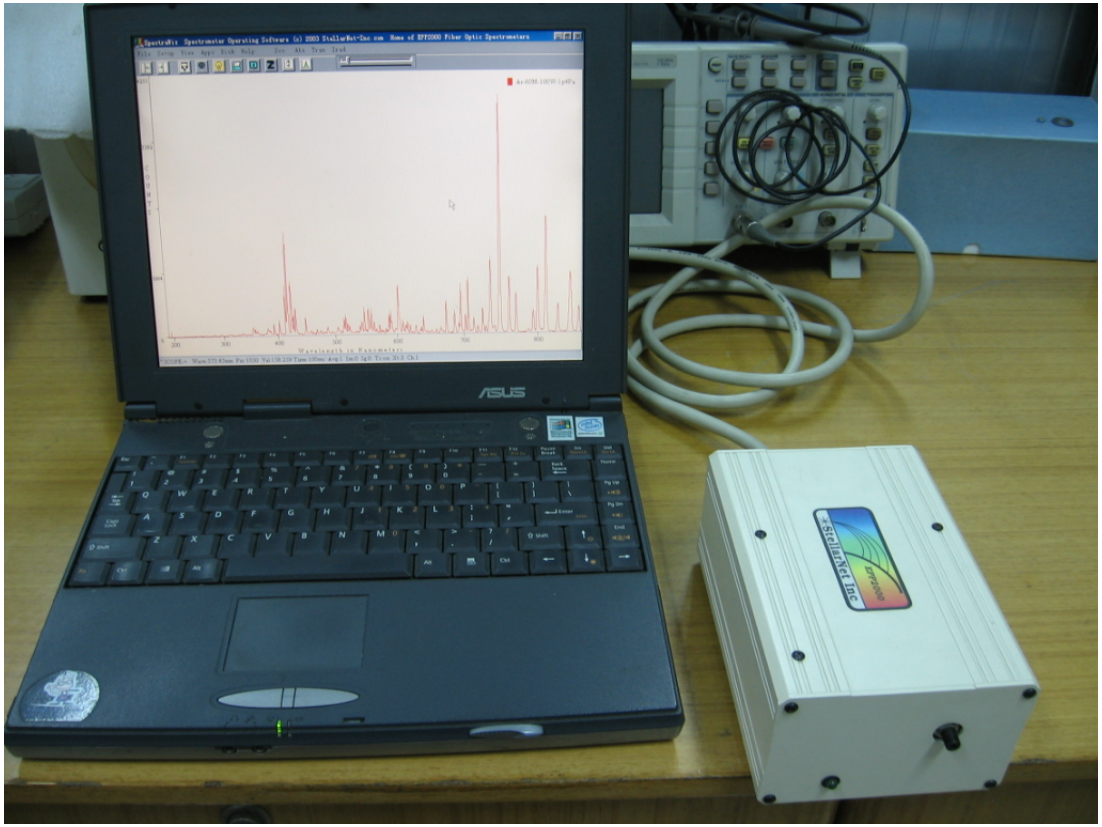
Electron Cyclotron Resonance Discharges with 2.45 GHz Microwave



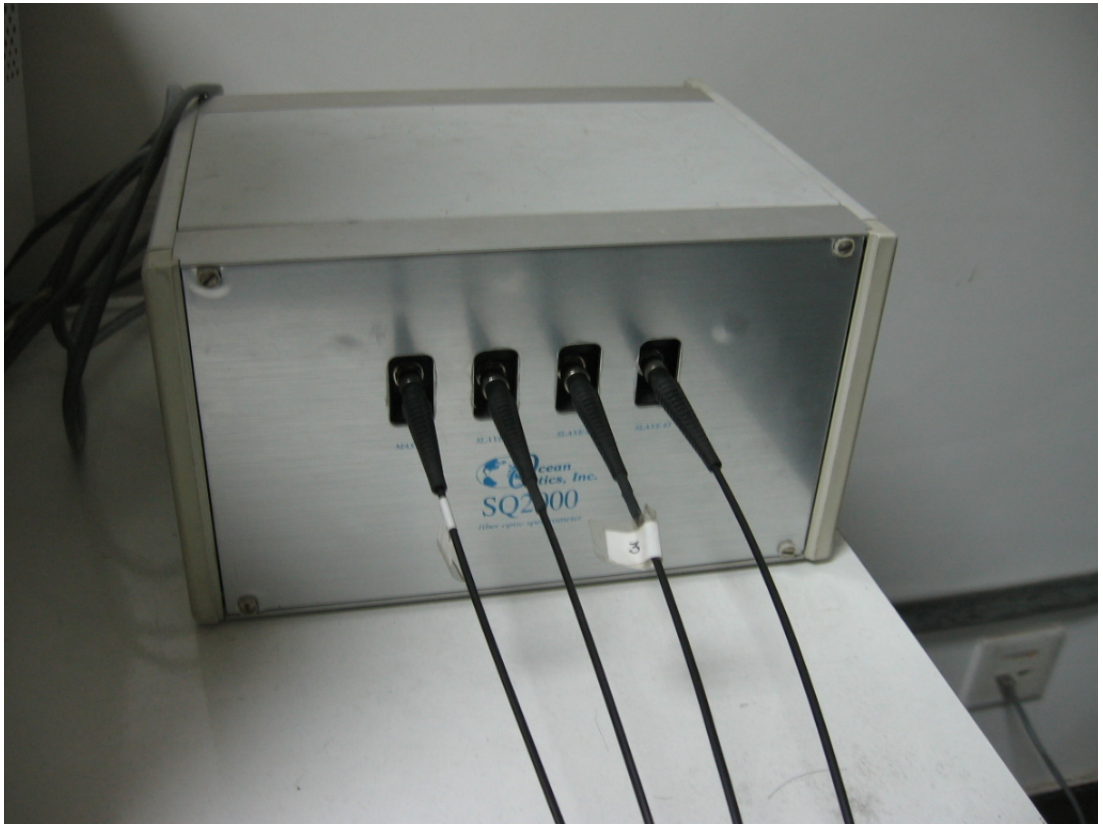
Multiple Frequency Capacitive Discharges with Ion Energy Analyzer and Mass Spectrometer



Portable CCD spectrometer



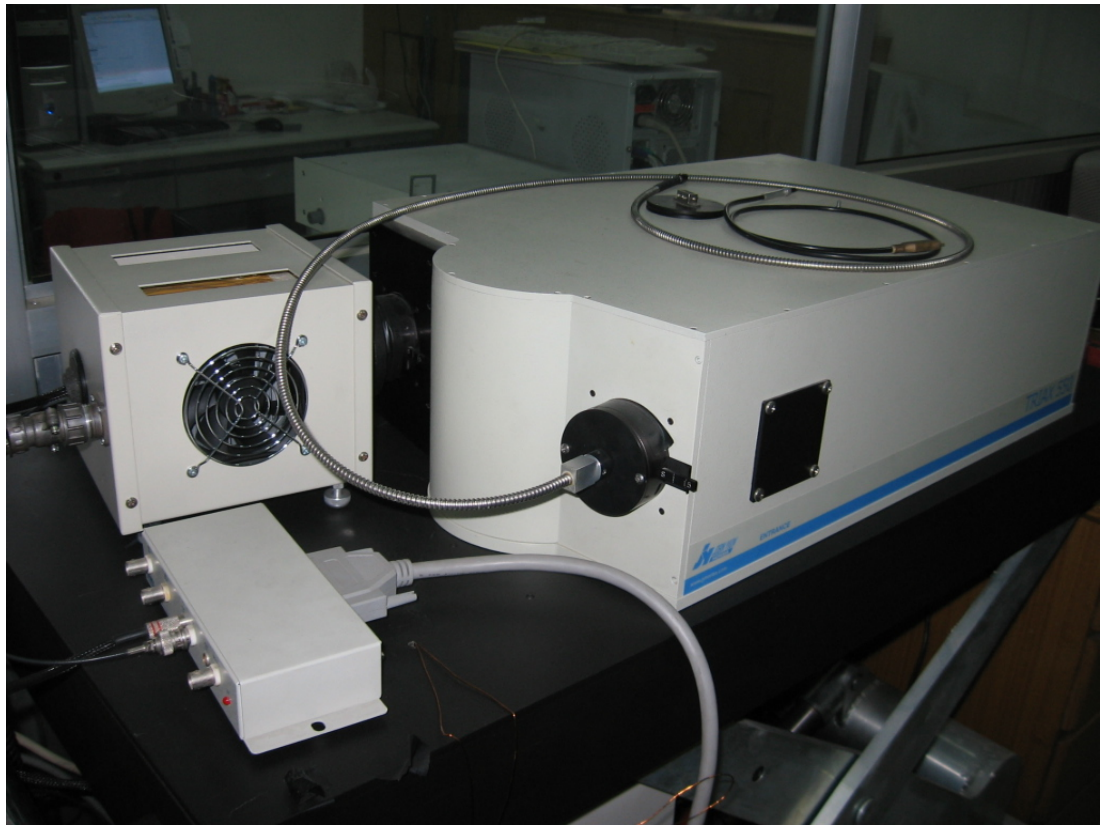
Four Channel CCD Spectrometer by Ocean Optics



High Resolution (0.15 nm) TE Cooled CCD Miniature Spectrometer



0.01 nm Resolution Spectrometer (made by Jobin-Yvon)



Some Recent Relevant Publications

- Comparative characterization of high density plasma reactors using emission spectroscopy from VUV to NIR, *Pure and Applied Chemistry*, **74**(3), pp.459~464, 2002.
- Tuning the electron temperature of a nitrogen plasma by adding helium and argon, Physics of Plasmas, v 10, n 10, Oct. 2003, p 4118-22
- Optical emission spectroscopy study of the influence of nitrogen on carbon nanotube growth, Carbon, v 41, n 9, 2003, p 1827-31
- Density measurement of helium metastable states by absorption spectroscopy in an inductively coupled plasma, Plasma Processes and Polymers, Vol.2, 581 (2005)
- Optical interference phenomenon in inductively coupled plasmas，J. Phys. D: Appl. Phys. 38 (2005) 3457-3460
- Observation of the emission spectra of an atmospheric pressure radio frequency plasma jet, Plasma Science & Technology, Vol. 7. No.5 3045 (2005)
- Discharge characteristics of an atmospheric pressure radio-frequency plasma jet, Journal of Physics D: Applied Physics, Vol. 38., 1396-1401, 2005
- Determining the electron temperature in inductively coupled nitrogen plasmas by optical emission spectroscopy with molecular kinetic effects, *PHYSICS OF PLASMAS*, **12**: 103501, 2005.
- Effect of wall reflection on the determination of electron temperature by the line-ratio method in inductively coupled plasmas, *PHYSICS OF PLASMAS*, 12, 113301, 2005
- Tuning effect of inert gas mixing on electron energy distribution function in inductively coupled, *Plasma Physics and Controlled Fusion*, Vol. 48, 61(2006)
- A molecular kinetic model for the OES technique in nitrogen ICP, *PHYSICS OF PLASMAS*, 2006.
- A novel method to determine electron density by optical emission spectroscopy in low-pressure nitrogen plasmas，PHYSICS OF PLASMAS, 13，123501, 2006.
- A simple collisional-radiative model for low-pressure argon discharges，JOURNAL OF PHYSICS D-APPLIED PHYSICS, 2007.